



Dendro-Pi Monitoring System

This project sets up a Raspberry Pi to monitor dendrometers and take periodic images, then uploads data to Dropbox for long-term analysis at research sites.

My other projects that build on this initial setup:

- [Charge Controller](#)
- Dendrometer Logger - *Under construction...*

The default behaviour of this system is to take a picture at 9:00 AM, 12:00 PM, 3:00 PM, and 6:00 PM every day, and upload the pictures each night to a Dropbox folder named after the Pi's hostname (e.g., [Dorval-8](#)).



Initial Setup: Flashing the Pi SD Card

1. Insert SD card into your computer.
 2. Download and open [Raspberry Pi Imager](#).
 3. Configure with the following:
 - **Device:** Raspberry Pi Zero (**Not Zero 2 W**)
 - **OS:** Raspberry Pi OS 32-bit (Bookworm)
 - **Storage:** Select Storage (Mass Storage Device USB Device)

- Edit settings

 - **Hostname:** e.g., [Dorval-8](#)
 - **Username/Password** (To log into the Pi): [madlab](#) / _____
 - **Wi-Fi SSID/Password:** [new_aspen_2022](#) / _____
 - **Country:** CA
 - **Enable SSH:** Use password authentication
 4. Save
 - Apply OS Customization Settings: Yes
 5. When done, insert the SD card into your Pi and power it up.
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Connect & Configure the Raspberry Pi

1. Connect your laptop to the same Wi-Fi as the Pi.
2. Open terminal (or PowerShell) and SSH into your Pi:

```
ssh madlab@Dorval-8.local # ssh username@{hostname}.local
```

if cannot resolve the hostname, use the IP address:

```
nslookup Dorval-8.local # to find the IP address
ssh madlab@{IP_ADDRESS} # ssh username@{IP_ADDRESS}
```

3. Run:

```
sudo raspi-config
```

- Set timezone under Localization Options
- Reboot the Pi

4. After reboot, SSH in again and test the camera:

```
rpicam-jpeg -o test.jpg
```

- If not found, install:

```
sudo apt-get update
sudo apt-get install python-picamera2 python3-picamera2
```

Installing dendro-pi Scripts

1. Clone the project:

```
git clone https://github.com/alanfogel/dendro-pi-main.git
cd dendro-pi-main
```

2. Enter test folder and try:

```
cd test
python test_camera.py
```

3. Edit your camera name in main/dendro_pictures.py on this line: `CAMERA_NAME = "DorvalTest"`

- (Replace `Dorval-8_` with your camera name):

```
cd ..
nano main/dendro_pictures.py
```

```
CAMERA_NAME = "Dorval-8_" # the trailing underscore is important for naming the files
```

☁ Configure Dropbox Upload

1. In the root project directory `~/dendro-pi-main/`:

```
git clone https://github.com/alanfogel/Dropbox-Uploader.git
cd Dropbox-Uploader
sudo chmod +x dropbox_uploader.sh
```

```
./dropbox_uploader.sh
```

- At the end of the output of the above command, it will ask for the "App key"
- Go to the Dropbox app developer url and login with your Dropbox account, a screen will appear having a "Create an app" button, and a list of Apps created with your account. Select "DendroPictures"
- Under the settings tab - Scroll down and you will find the "App key" and "App secret", note down them and return back to the terminal
- In the terminal enter the codes when prompted, (when you enter the "App secret", then it will give you a link, visiting it, you will get the "Access code"), once all the information is provided you will link with your dropbox cloud.

2. Update upload-to-dropbox.sh:

```
cd ..
nano upload-to-dropbox.sh
```

Modify the variable declaration line (Replace `Dorval-8` with your Dropbox folder name):

```
DROPBOX_PATH="/Dorval-8/"
```

3. Ensure UNIX line endings:

```
dos2unix upload-to-dropbox.sh
```

🕒 Setup Crontab

1. Open crontab and save immediately:

```
crontab -e
```

2. Check if its empty:

```
crontab -l
```

3. If it's not empty, run:

```
crontab -r
```

4. Install scheduled jobs: **TODO: FIX ADD_CRON.sh**

```
sh add_cron.sh
```

5. Confirm:

```
crontab -l
```

6. You can edit the crontab file to change the schedule (like staggering the uploads):

```
crontab -e
```

✅ Test Setup

```
# Take a picture
cd main
python dendro_pictures.py

# Upload pictures
cd ..
bash upload-to-dropbox.sh
```

- Check Dropbox for uploaded files.

📁 Copying Pictures from Pi to PC

From your Windows machine:

```
scp -r madlab@Dorval8.local:~/dendro-pi-main/pictures C:\Users\alanj\Desktop
```

- Replace `Dorval8.local` with your Pi's hostname.
- Replace `C:\Users\alanj\Desktop` with your desired local directory.